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Dear Dr Kousta,

We would like to submit the attached manuscript for your consideration as a regular research article at *Nature Human Behaviour*.

Registered Reports are widely seen as the most promising remedy for publication bias. Part of the case for the format is that it also serves the interests of individual researchers by removing the risk of not being able to publish a study due to unfavourable results.

Our paper addresses the question of whether this factor is actually likely to make Registered Reports universally attractive to researchers. We assume that authors care not only about their study being published at all, but also about journal impact, and that outside of the Registered Reports format, publication in high-impact journals depends at least in part on the study results. Under these assumptions – and because most high-impact journals do not yet offer the format (*NHB* being a notable exception) – Registered Reports' pre-data publication guarantee caps not only the worst possible outcome for authors (no publication), but also the best (publication in a very high-impact journal). Therefore, authors' choice between publication formats can be characterised as decision-making under risk. We borrow risk-sensitivity theory from behavioural ecology and develop an agent-based model to explore how four factors of academic careers may shape researchers' publication preferences: whether additional publications yield diminishing or increasing career returns, empirical pace, publication targets, and competition. Our results suggest that Registered Reports may be surprisingly unpopular under many realistic constellations, especially when researchers must compete for jobs or grant funding. We discuss the implications of these findings and possible interventions to make Registered Reports more attractive.

All simulation code, simulated data, and the materials for reproducing the figures are available online (github.com/amscheel/rr-model). A first version of the manuscript, with identical full text but a slightly longer abstract, has been uploaded as a preprint to *bioRxiv*. The manuscript has not been submitted to any other platform or journal.

Thank you very much for your consideration.

Kind regards,

Anne Scheel

on behalf of my co-authors Leo Tiokhin and Daniël Lakens